Ahsanullah University of Science and Technology



Department of Electrical & Electronic Engineering

DESIGN PROJECT: POWER FLOW

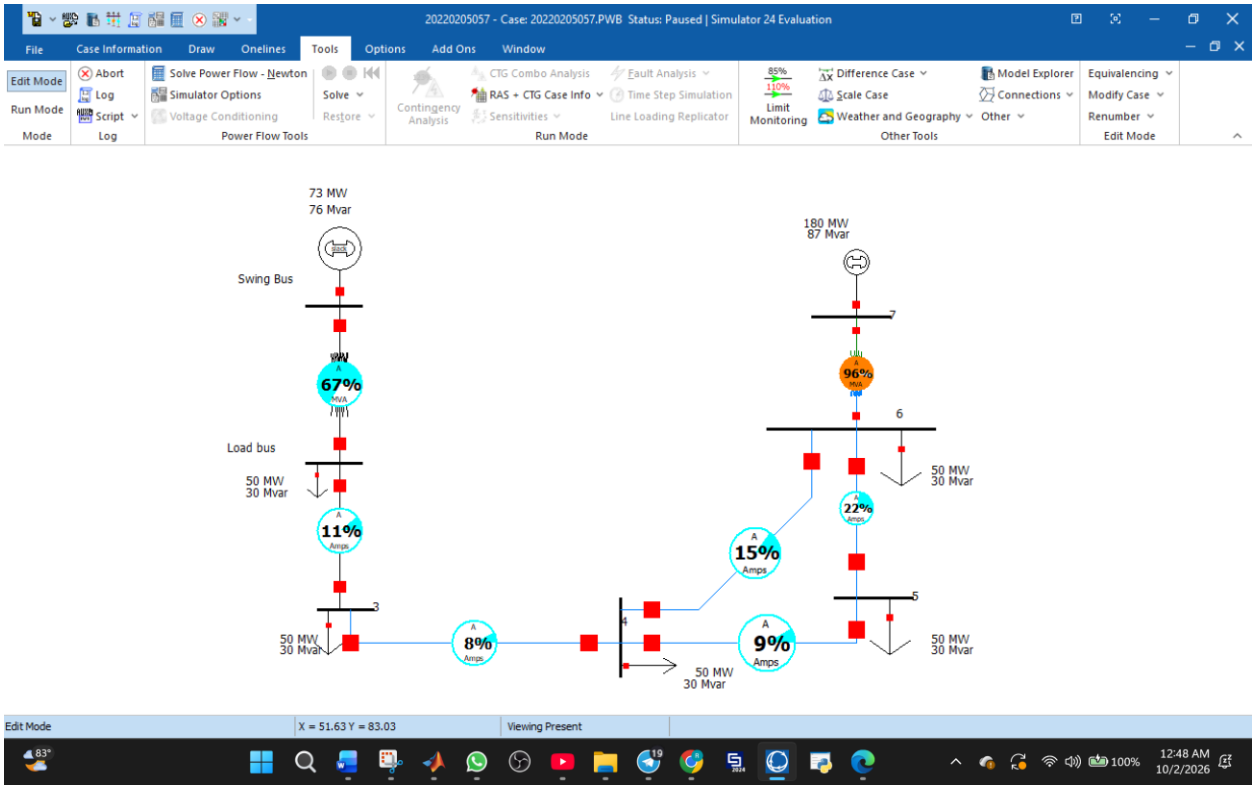
Course No	: **EEE 4154**
Course Name	: **Power System II Lab**
Date of Submission	: 2/10/2026
Experiment no	: 9 & 10

Submitted By:

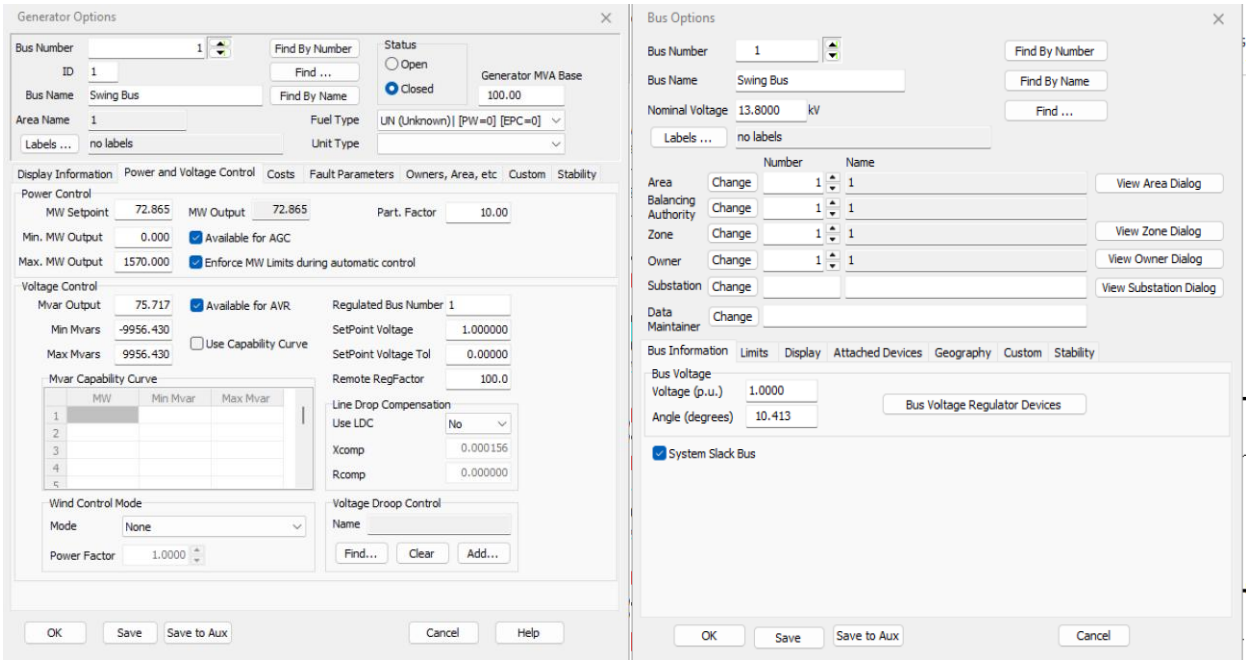
Id	: 20220205057
Section	: A (A2)
Year	: 4th
Semester	: 1st
Group	: 4

Objective: The objective of this experiment is to analyze the power flow in a given power system using the PowerWorld Simulator, and design a different operating strategy to change certain system variables.

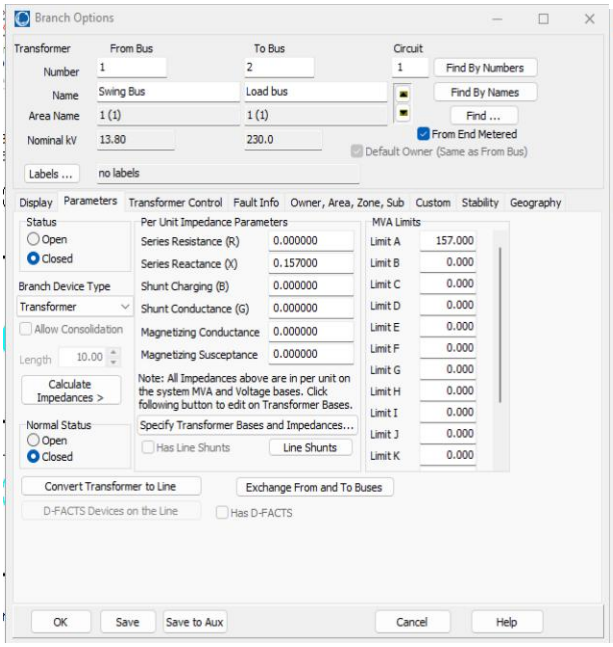
Power World Simulation:

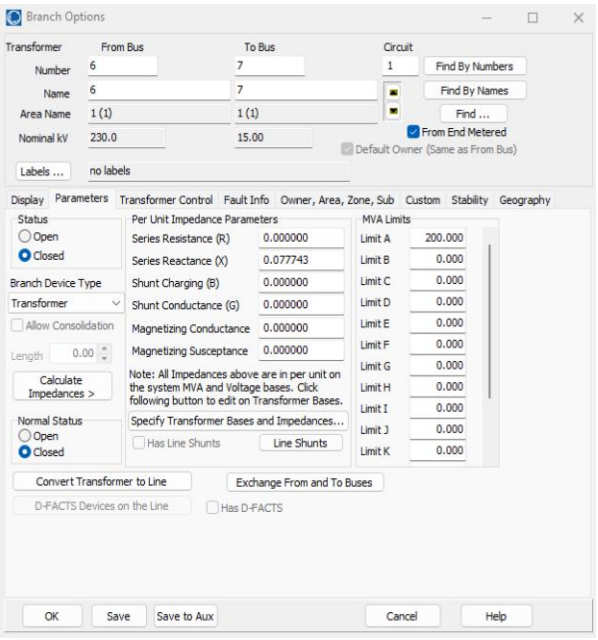


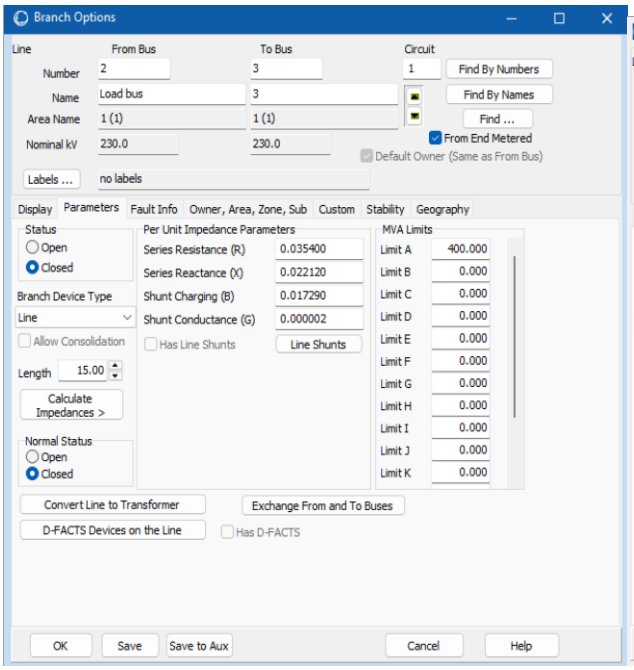
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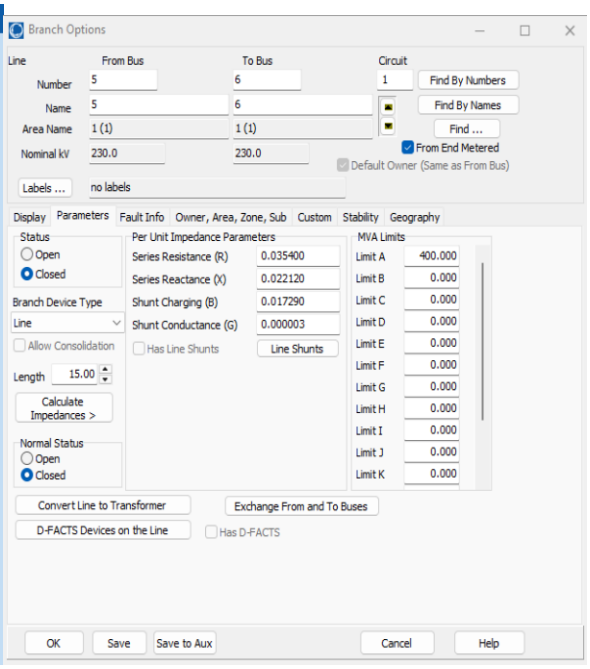


Transmission Lines:

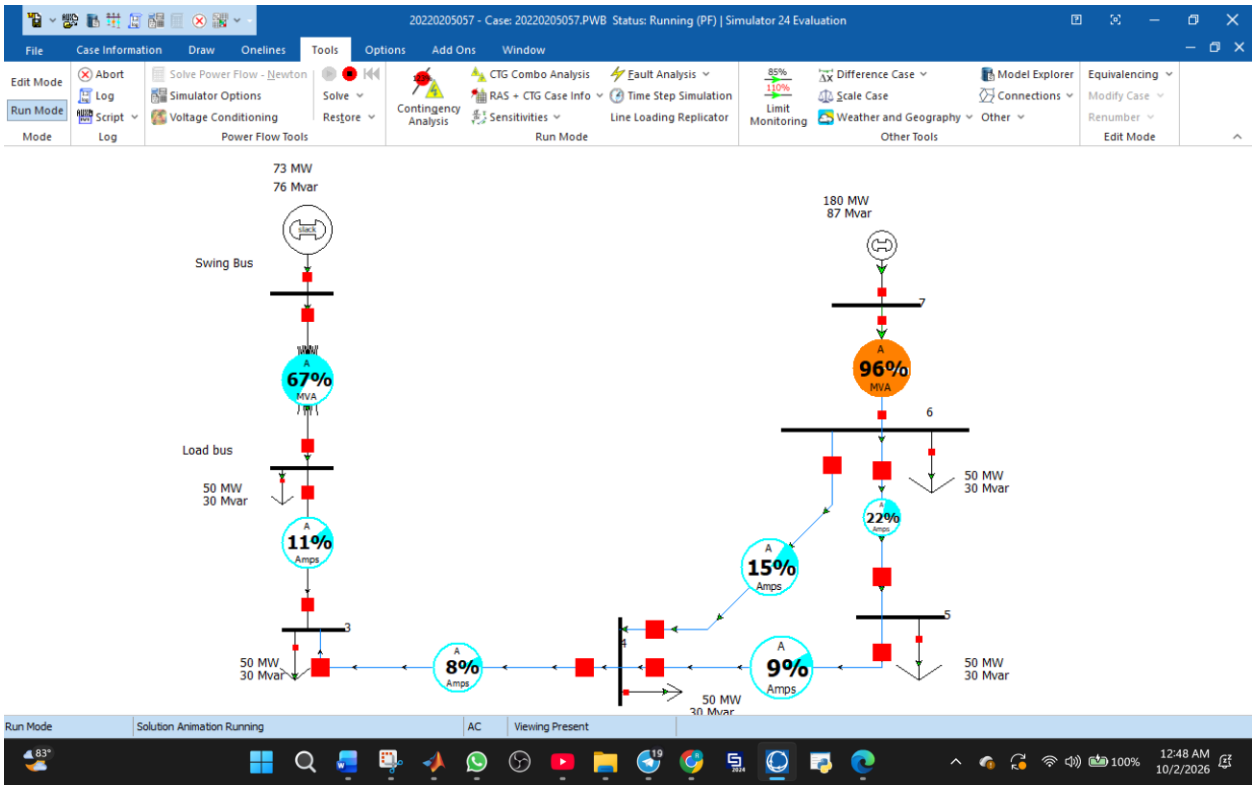

Branch Options
Transformer
From Bus
To Bus
Circuit
Number 1 2 1
Name Swing Bus Load bus
Area Name 1 (1) 1 (1)
Nominal kV 13.80 230.0
Find By Numbers
Find By Names
Find ...
From End Metered
Default Owner (Same as From Bus)
Labels ... no labels
Display Parameters Transformer Control Fault Info Owner, Area, Zone, Sub Custom Stability Geography
Status
Open
Closed
Branch Device Type
Transformer
Allow Consolidation
Length 10.00
Calculate Impedances >
Normal Status
Open
Closed
Per Unit Impedance Parameters
Series Resistance (R) 0.000000
Series Reactance (X) 0.157000
Shunt Charging (B) 0.000000
Shunt Conductance (G) 0.000000
Magnetizing Conductance 0.000000
Magnetizing Susceptance 0.000000
Note: All Impedances above are in per unit on the system MVA and Voltage bases. Click following button to edit on Transformer Bases.
Specify Transformer Bases and Impedances...
Has Line Shunts
Line Shunts
MVA Limits
Limit A 157.000
Limit B 0.000
Limit C 0.000
Limit D 0.000
Limit E 0.000
Limit F 0.000
Limit G 0.000
Limit H 0.000
Limit I 0.000
Limit J 0.000
Limit K 0.000
Convert Transformer to Line
Exchange From and To Buses
D-FACTS Devices on the Line
Has D-FACTS
OK
Save
Save to Aux
Cancel
Help


Branch Options
Transformer
From Bus
To Bus
Circuit
Number 6 7 1
Name 6 7
Area Name 1 (1) 1 (1)
Nominal kV 230.0 15.00
Find By Numbers
Find By Names
Find ...
From End Metered
Default Owner (Same as From Bus)
Labels ... no labels
Display Parameters Transformer Control Fault Info Owner, Area, Zone, Sub Custom Stability Geography
Status
Open
Closed
Branch Device Type
Transformer
Allow Consolidation
Length 0.00
Calculate Impedances >
Normal Status
Open
Closed
Per Unit Impedance Parameters
Series Resistance (R) 0.000000
Series Reactance (X) 0.077743
Shunt Charging (B) 0.000000
Shunt Conductance (G) 0.000000
Magnetizing Conductance 0.000000
Magnetizing Susceptance 0.000000
Note: All Impedances above are in per unit on the system MVA and Voltage bases. Click following button to edit on Transformer Bases.
Specify Transformer Bases and Impedances...
Has Line Shunts
Line Shunts
MVA Limits
Limit A 200.000
Limit B 0.000
Limit C 0.000
Limit D 0.000
Limit E 0.000
Limit F 0.000
Limit G 0.000
Limit H 0.000
Limit I 0.000
Limit J 0.000
Limit K 0.000
Convert Transformer to Line
Exchange From and To Buses
D-FACTS Devices on the Line
Has D-FACTS
OK
Save
Save to Aux
Cancel
Help

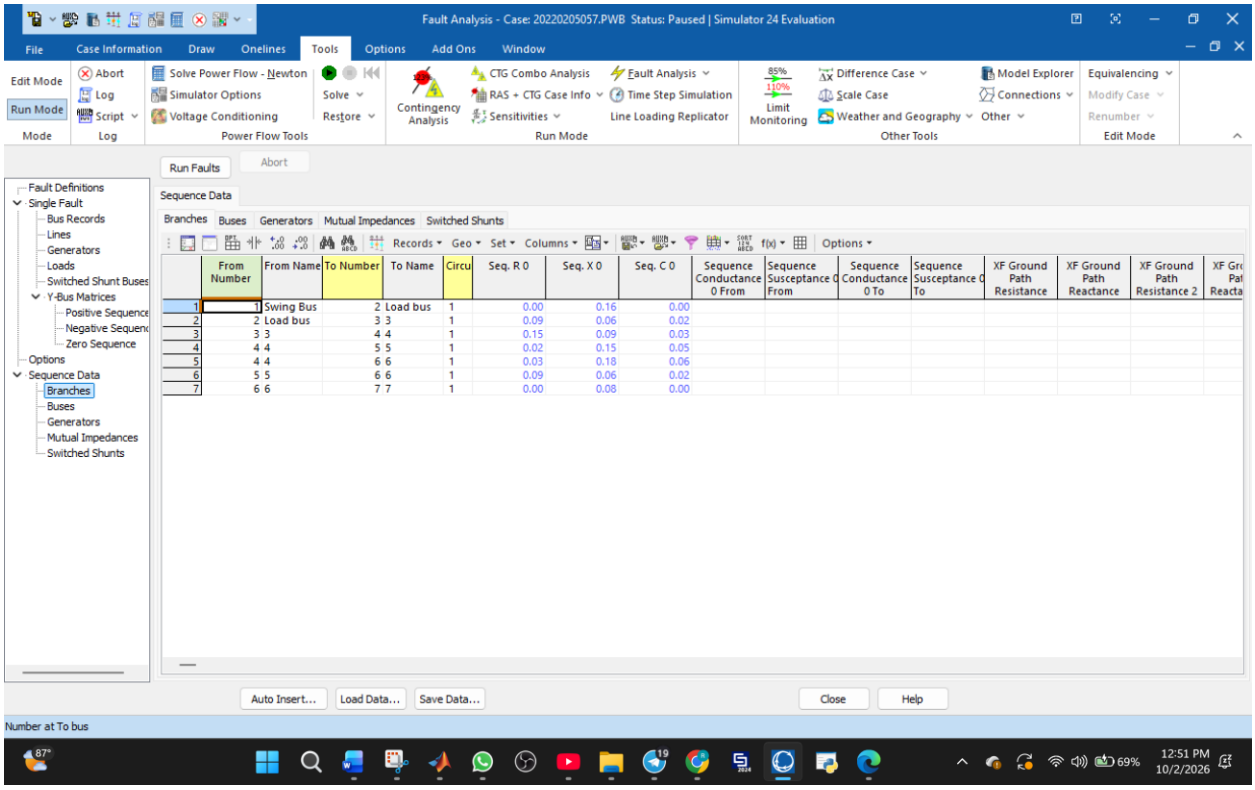

Branch Options
Line
From Bus
To Bus
Circuit
Number 2 3 1
Name Load bus 3
Area Name 1 (1) 1 (1)
Nominal kV 230.0 230.0
Find By Numbers
Find By Names
Find ...
From End Metered
Default Owner (Same as From Bus)
Labels ... no labels
Display Parameters Fault Info Owner, Area, Zone, Sub Custom Stability Geography
Status
Open
Closed
Branch Device Type
Line
Allow Consolidation
Length 15.00
Calculate Impedances >
Normal Status
Open
Closed
Per Unit Impedance Parameters
Series Resistance (R) 0.035400
Series Reactance (X) 0.022120
Shunt Charging (B) 0.017290
Shunt Conductance (G) 0.000002
Has Line Shunts
Line Shunts
MVA Limits
Limit A 400.000
Limit B 0.000
Limit C 0.000
Limit D 0.000
Limit E 0.000
Limit F 0.000
Limit G 0.000
Limit H 0.000
Limit I 0.000
Limit J 0.000
Limit K 0.000
Convert Line to Transformer
Exchange From and To Buses
D-FACTS Devices on the Line
Has D-FACTS
OK
Save
Save to Aux
Cancel
Help

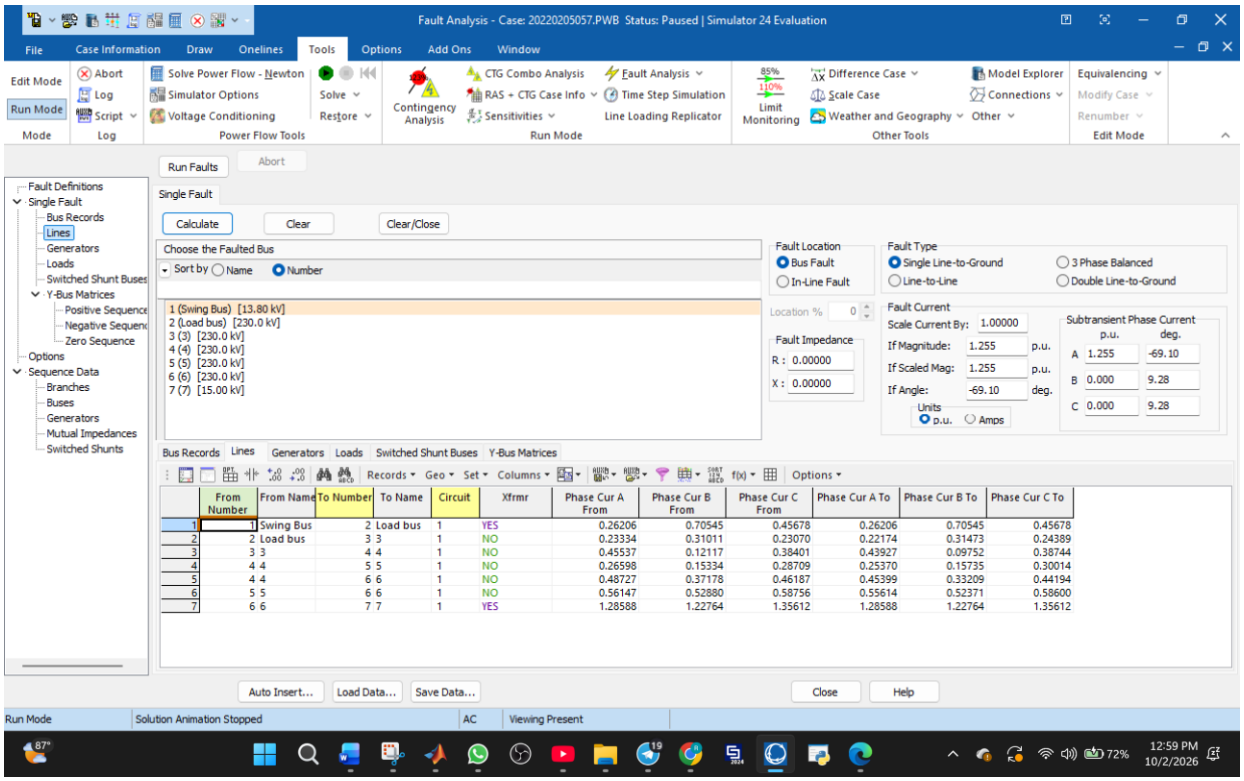

Branch Options
Line
From Bus
To Bus
Circuit
Number 5 6 1
Name 5 6
Area Name 1 (1) 1 (1)
Nominal kV 230.0 230.0
Find By Numbers
Find By Names
Find ...
From End Metered
Default Owner (Same as From Bus)
Labels ... no labels
Display Parameters Fault Info Owner, Area, Zone, Sub Custom Stability Geography
Status
Open
Closed
Branch Device Type
Line
Allow Consolidation
Length 15.00
Calculate Impedances >
Normal Status
Open
Closed
Per Unit Impedance Parameters
Series Resistance (R) 0.035400
Series Reactance (X) 0.022120
Shunt Charging (B) 0.017290
Shunt Conductance (G) 0.000003
Has Line Shunts
Line Shunts
MVA Limits
Limit A 400.000
Limit B 0.000
Limit C 0.000
Limit D 0.000
Limit E 0.000
Limit F 0.000
Limit G 0.000
Limit H 0.000
Limit I 0.000
Limit J 0.000
Limit K 0.000
Convert Line to Transformer
Exchange From and To Buses
D-FACTS Devices on the Line
Has D-FACTS
OK
Save
Save to Aux
Cancel
Help

After Run:



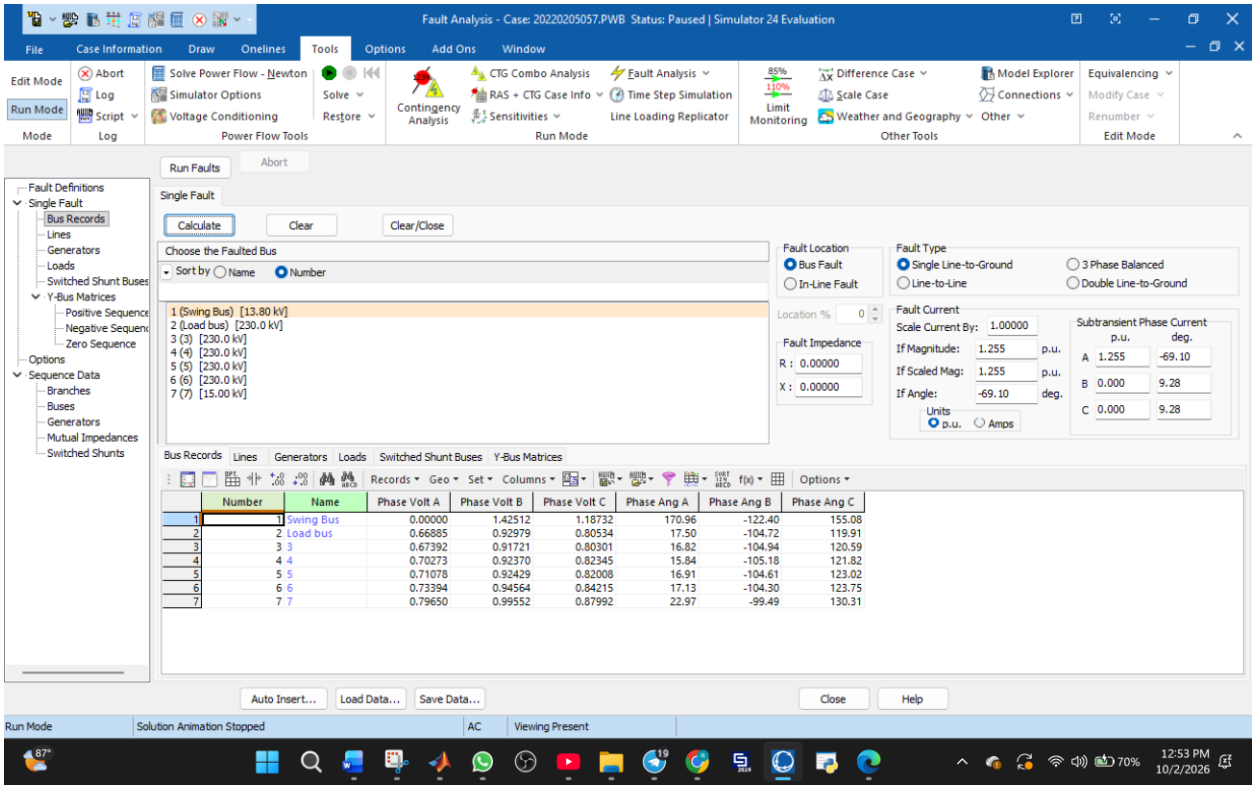
Branches Input:



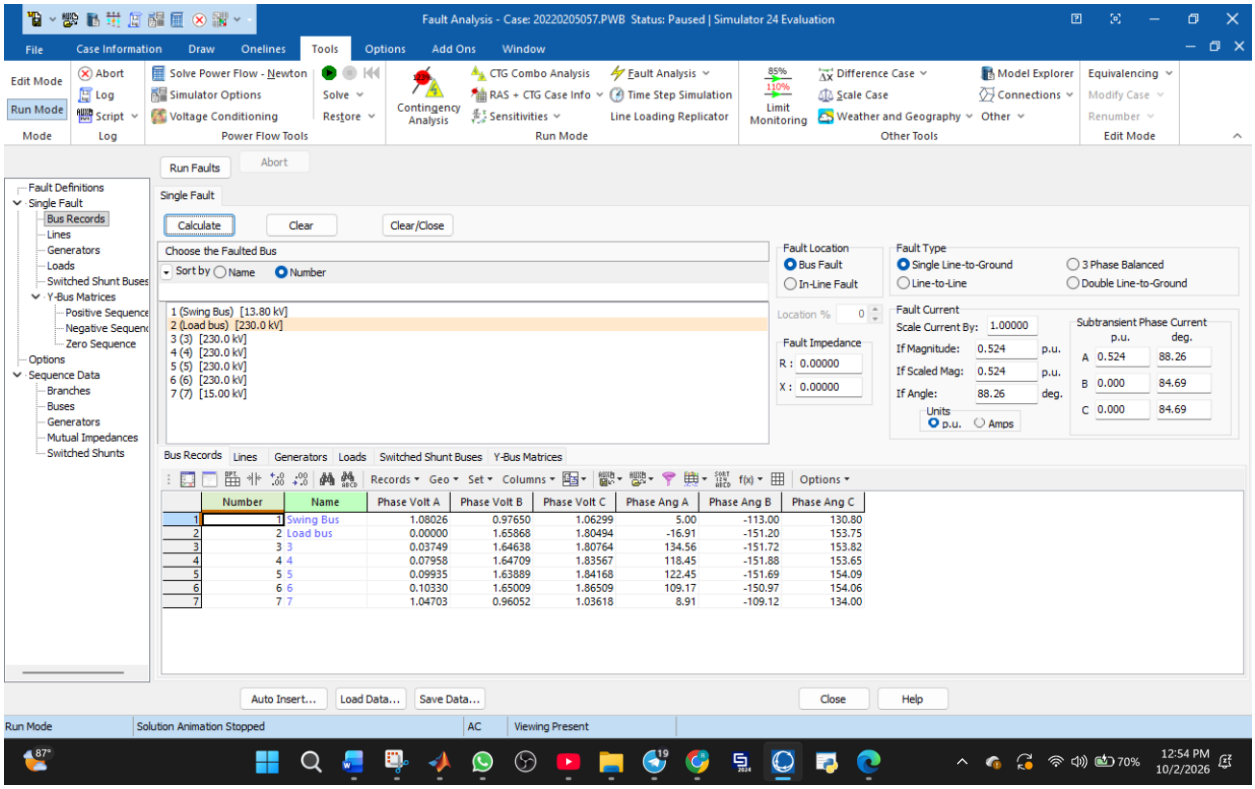


Bus Records:

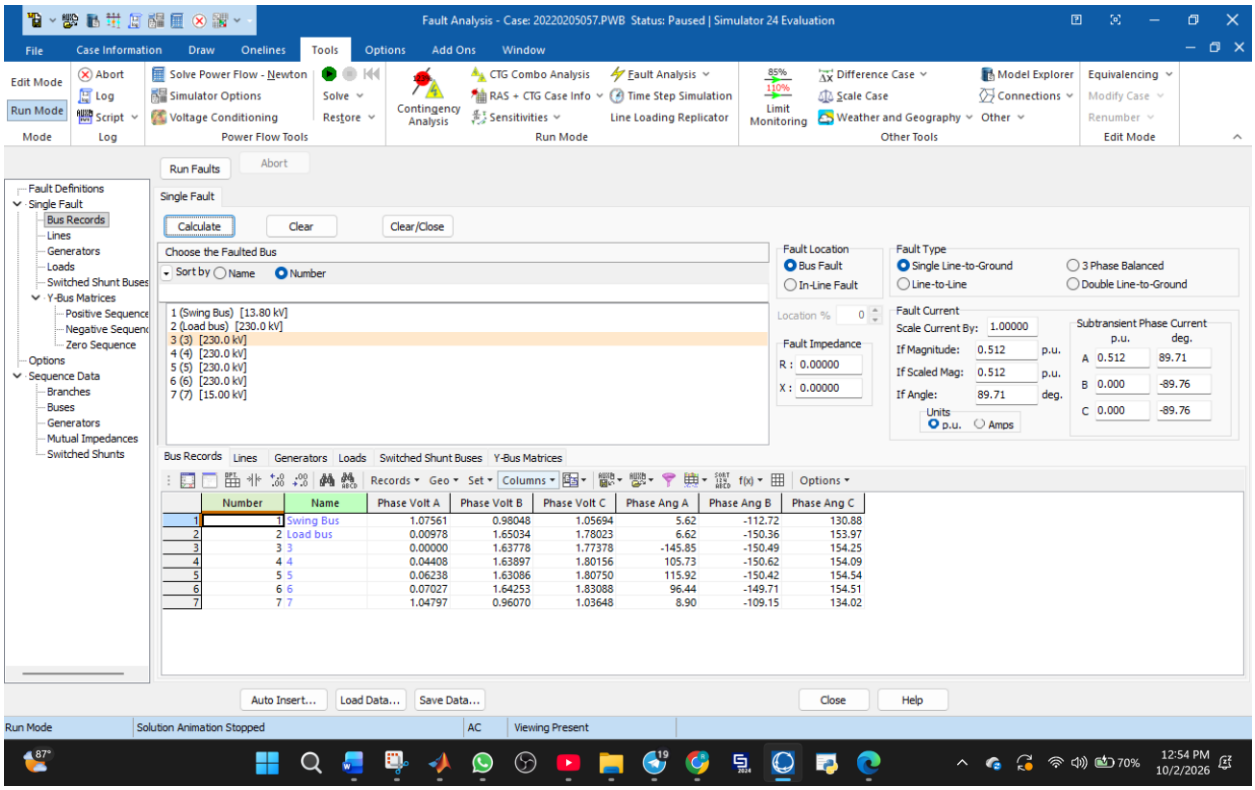
Swing Bus:



Load Bus:

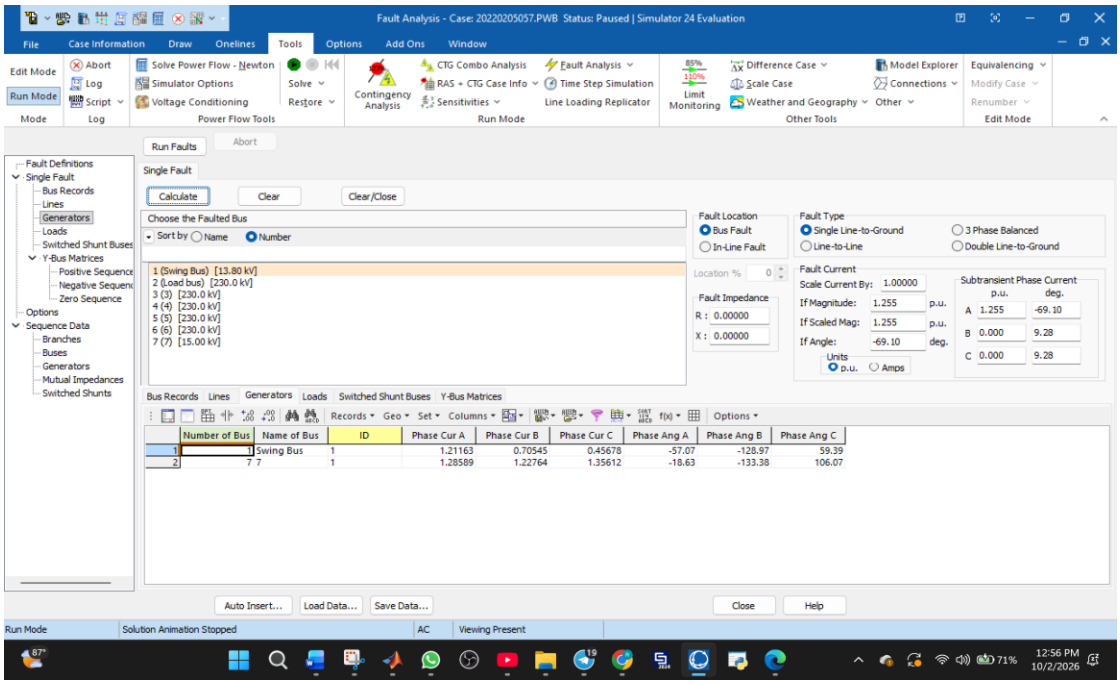


PV Bus:

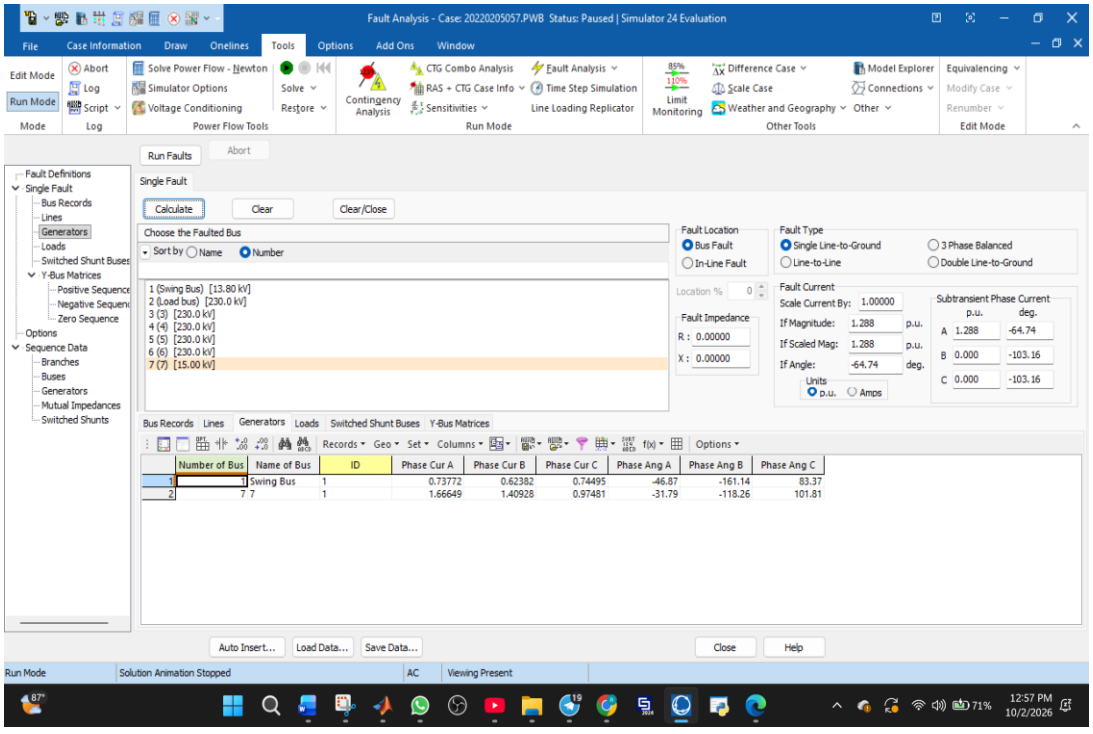


Generator Records:

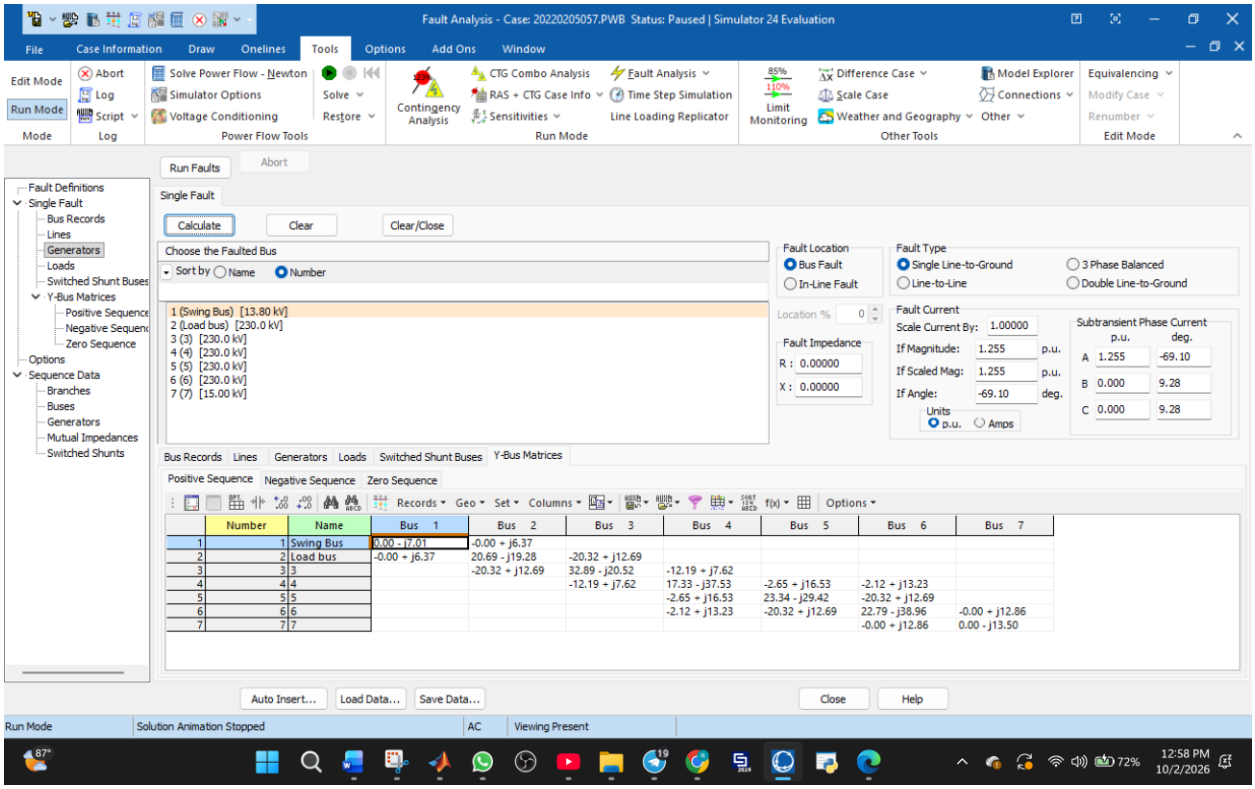
Gen1:



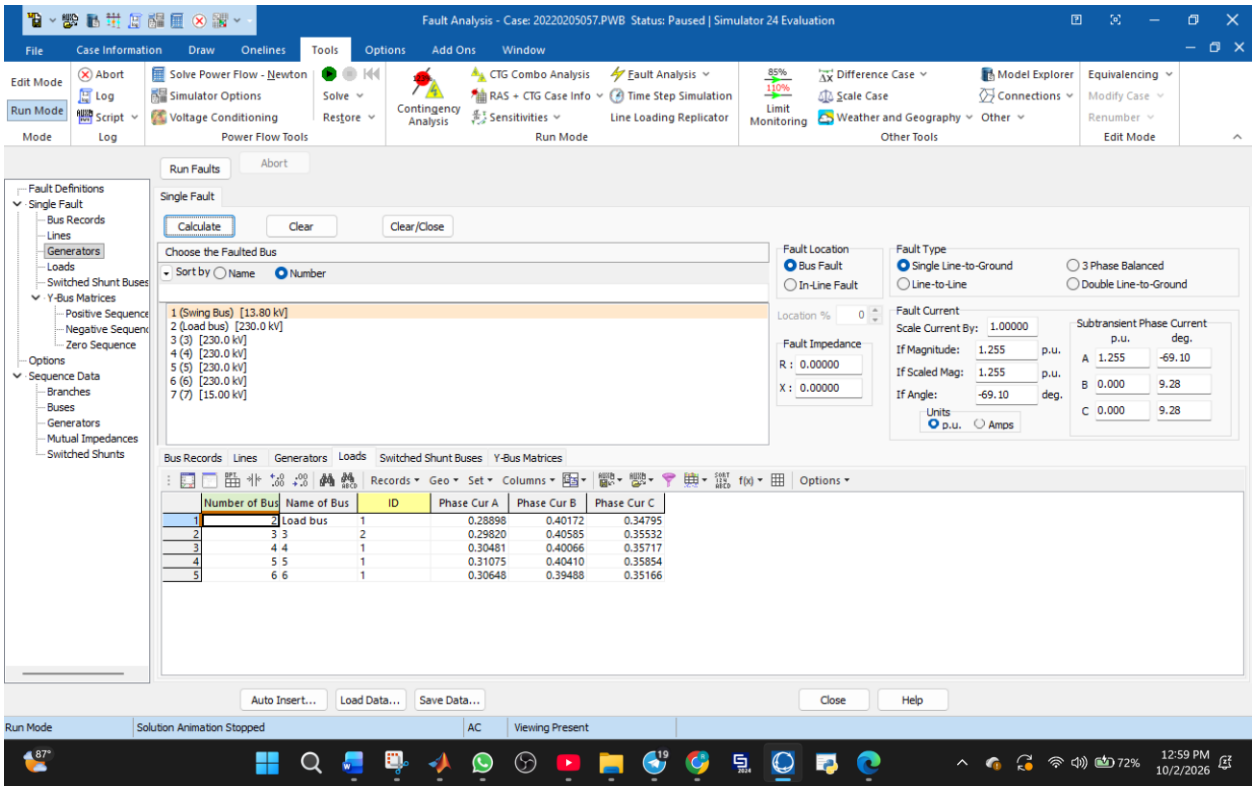
Gen 2:



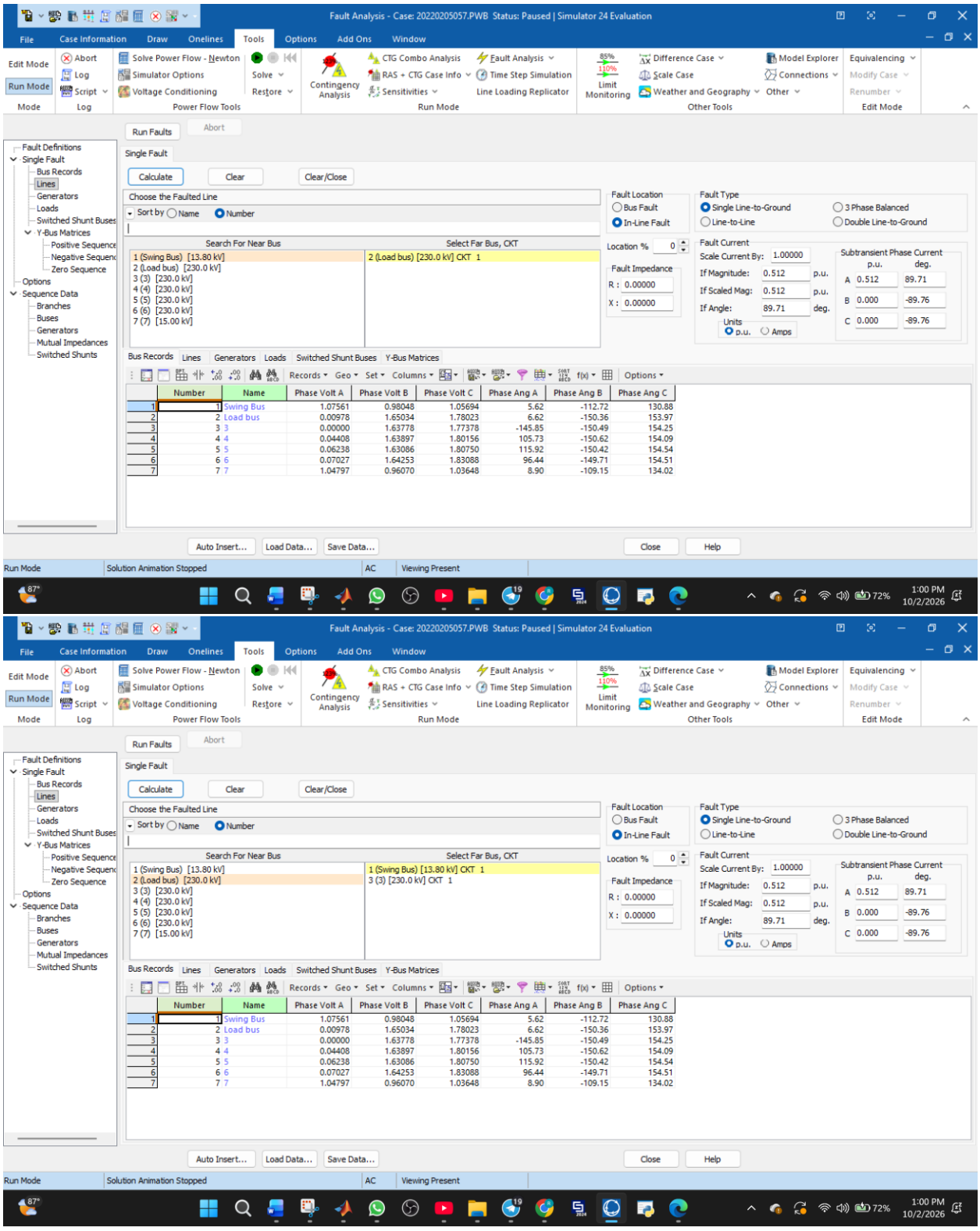
Y-Bus:


Fault Analysis - Case: 20220205057.PWB Status: Paused | Simulator 24 Evaluation
Run Faults
Single Fault
Calculate
Clear
Clear/Close
Choose the Faulted Bus
1 (Swing Bus) [13.80 kV]
2 (Load bus) [230.0 kV]
3 (3) [230.0 kV]
4 (4) [230.0 kV]
5 (5) [230.0 kV]
6 (6) [230.0 kV]
7 (7) [15.00 kV]
Fault Location
Bus Fault
In-Line Fault
Fault Type
Single Line-to-Ground
Line-to-Line
3 Phase Balanced
Double Line-to-Ground
Fault Impedance
R : 0.00000
X : 0.00000
Fault Current
Scale Current By: 1.00000
If Magnitude: 1.255 p.u.
If Scaled Mag: 1.255 p.u.
If Angle: -69.10 deg.
Subtransient Phase Current
A 1.255 -69.10
B 0.000 9.28
C 0.000 9.28
Bus Records
Lines
Generators
Loads
Switched Shunt Buses
Y-Bus Matrices
Positive Sequence
Negative Sequence
Zero Sequence
Number
Name
Bus 1
Bus 2
Bus 3
Bus 4
Bus 5
Bus 6
Bus 7
Swing Bus
Load bus
Auto Insert...
Load Data...
Save Data...
Close
Help
Run Mode
Solution Animation Stopped
AC
Viewing Present

Loads:

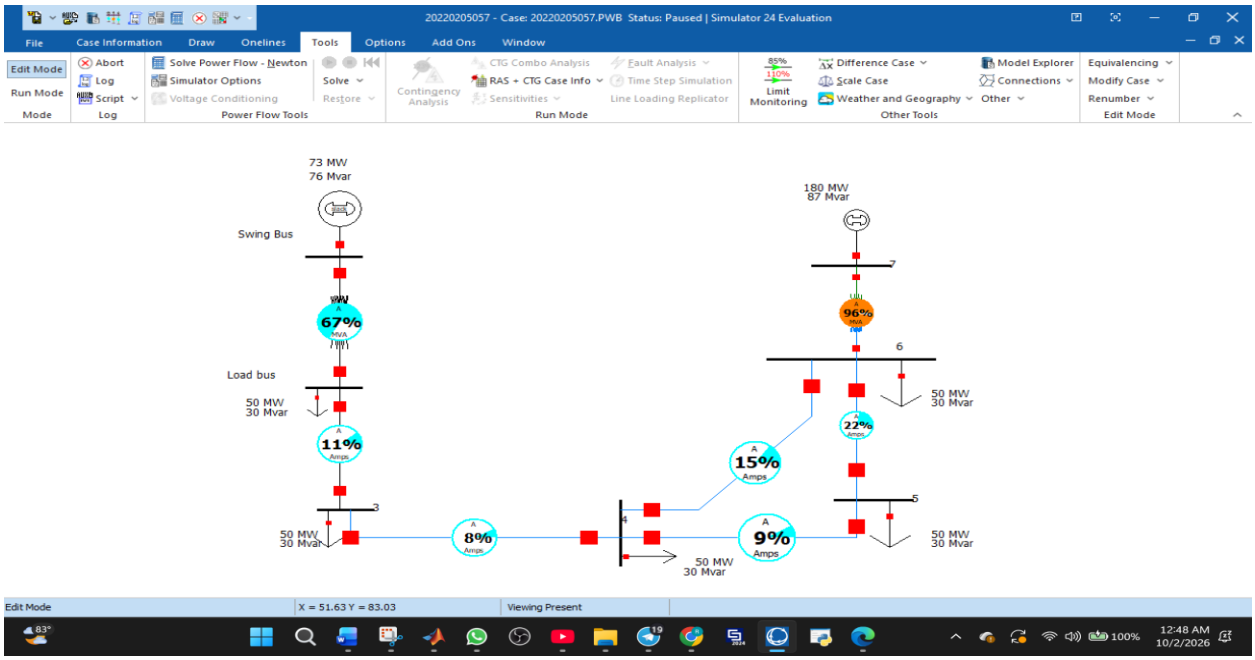

Fault Analysis - Case: 20220205057.PWB Status: Paused | Simulator 24 Evaluation
Run Faults
Single Fault
Calculate
Clear
Clear/Close
Choose the Faulted Bus
1 (Swing Bus) [13.80 kV]
2 (Load bus) [230.0 kV]
3 (3) [230.0 kV]
4 (4) [230.0 kV]
5 (5) [230.0 kV]
6 (6) [230.0 kV]
7 (7) [15.00 kV]
Fault Location
Bus Fault
In-Line Fault
Fault Type
Single Line-to-Ground
Line-to-Line
3 Phase Balanced
Double Line-to-Ground
Fault Impedance
R : 0.00000
X : 0.00000
Fault Current
Scale Current By: 1.00000
If Magnitude: 1.255 p.u.
If Scaled Mag: 1.255 p.u.
If Angle: -69.10 deg.
Subtransient Phase Current
A 1.255 -69.10
B 0.000 9.28
C 0.000 9.28
Bus Records
Lines
Generators
Loads
Switched Shunt Buses
Y-Bus Matrices
Number of Bus
Name of Bus
ID
Phase Cur A
Phase Cur B
Phase Cur C
Load bus
Auto Insert...
Load Data...
Save Data...
Close
Help
Run Mode
Solution Animation Stopped
AC
Viewing Present

In line Fault:

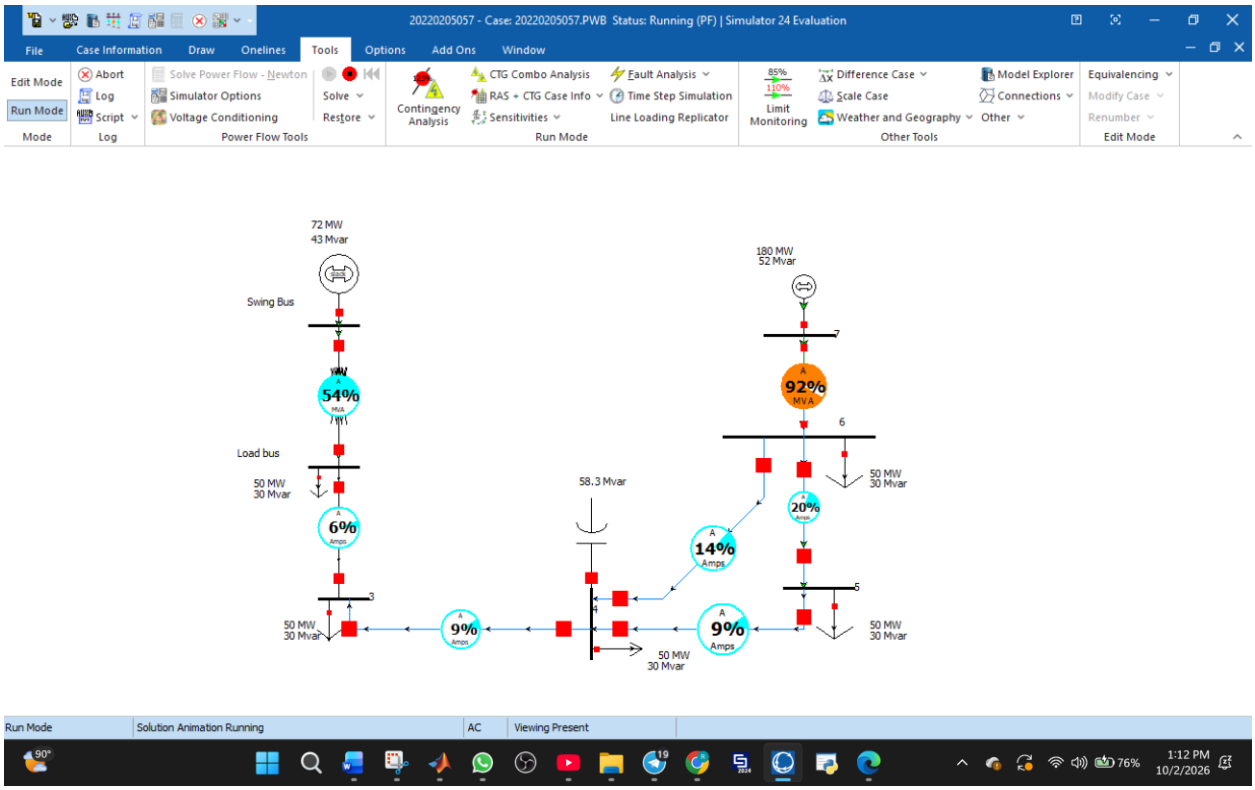


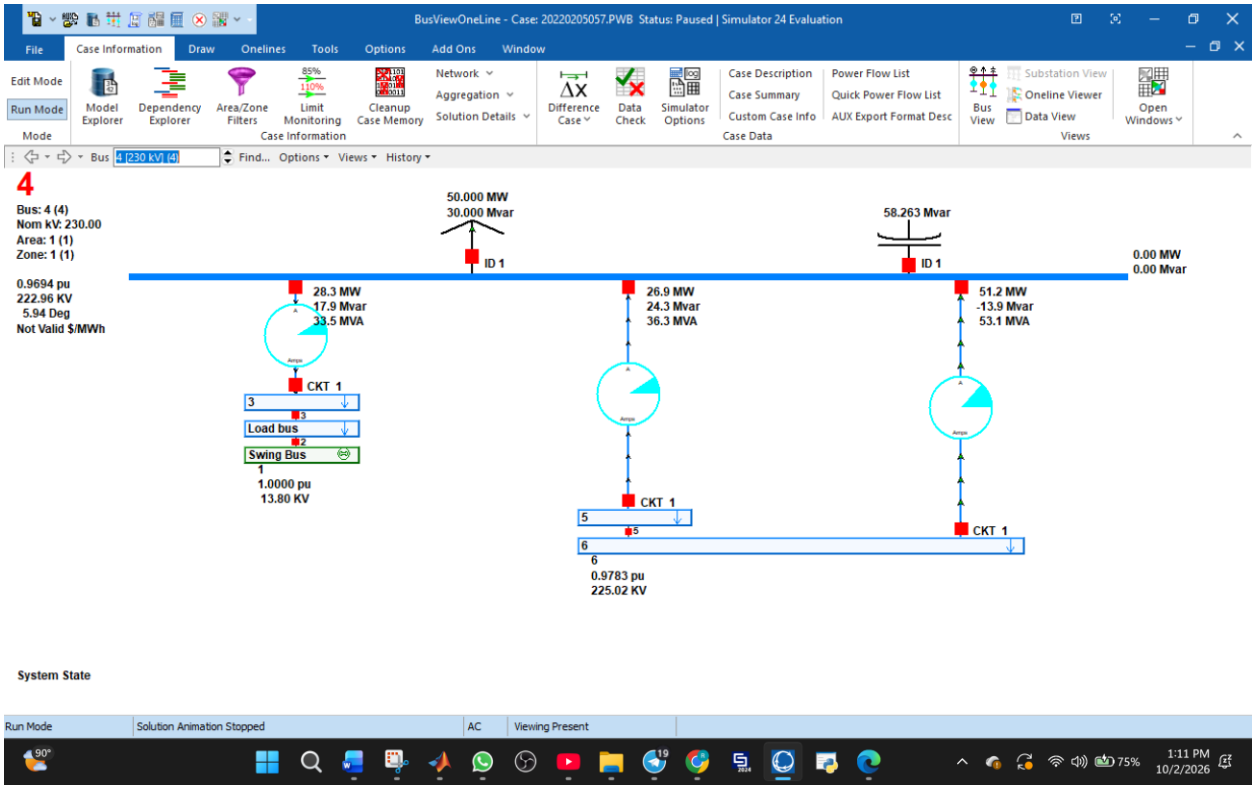
After Adding Shunt Capacitor:

Before:



After:





Conclusion

This project successfully demonstrated the complete workflow of power flow analysis using PowerWorld Simulator, from initial system modeling through to the design and verification of a corrective control strategy.

System Modeling and Per-Unit Conversion

The given 7-bus power system — comprising two generators (G1 at Bus 1, G2 at Bus 7), two transformers (T1, T2), five transmission lines (L1–L5), and five load buses — was first converted into per-unit quantities using a system base of **Sbase = 157 MVA**, derived from the assigned student ID (057). All generator reactances, transformer impedances, and transmission line parameters (resistance, reactance, and shunt susceptance) were recalculated onto this base before being entered into PowerWorld, ensuring the simulated case matched the hand-calculated values.

Model Verification and Troubleshooting

During model construction, several configuration errors were identified and corrected — including an incorrect bus nominal voltage, mismatched transformer reactances left over from base-value changes, and an unintended zero-value load — each of which initially drove the system into a non-converging "blackout" state. Systematically cross-checking the Bus, Branch, Generator, and Load data tables against the calculated per-unit values resolved these issues, after which the power flow solved successfully under the Newton-Raphson method without any equipment overload.

Base Case Results

The converged base case showed that Bus 4 — being electrically the most remote point in the network, fed through two long transmission lines and a meshed branch (L5) — experienced the most significant voltage drop in the system, with a solved voltage of only **0.9253 pu (212.83 kV)**, well below the nominal 230 kV level. This highlighted Bus 4 as the critical bus requiring reactive power support.

Voltage Improvement Strategy

To address this, a **static shunt capacitor bank** was selected as the corrective device, based on its simplicity, low cost, and ability to provide direct, localized reactive power support exactly where the voltage deficiency occurred — in contrast to generator excitation control or transformer tap-changing, which affect the network more broadly and are subject to additional operating limits. Through iterative sizing and re-solution of the power flow, a capacitor bank of approximately **58.3 Mvar** installed at Bus 4 raised its voltage from 212.83 kV to **222.96 kV**, an improvement of **4.76%**, closely achieving the targeted 5% voltage rise.

Overall Assessment

The results confirm that shunt capacitor compensation is an effective and economical method for correcting localized undervoltage conditions in a transmission network, particularly at buses that are electrically distant from

generation sources. While the achieved improvement (4.76%) fell marginally short of the exact 5% target, this gap could be closed with a modest increase in capacitor size (to roughly 62–63 Mvar), and the relationship between capacitor size and voltage rise observed during testing was found to be approximately linear over the tested range, making future fine-tuning straightforward. Compared to alternative solutions — new parallel transmission lines, SVCs, or STATCOMs — the shunt capacitor offers a substantially lower-cost path to achieving the required voltage support, making it the most practical choice for this application, though it should be noted that a fixed capacitor bank provides compensation suited to this specific loading condition and would need to be reassessed or supplemented with switched/dynamic compensation if load levels at Bus 4 vary significantly in practice.